YamScan: Automated Purple Yam Quality Detection and Classification Using a Two-Stage EfficientNetB0 Pipeline

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***Abstract*—Purple yam (*Dioscorea alata*), locally known as ube, is an important root crop in the Philippines and is increasingly used in food products and trade [1], [2]. Its quality grading is commonly conducted by visual inspection, a process that can be labor-intensive and subject to evaluator variation. This paper presents YamScan, a computer-vision system that first identifies yam images and then assigns detected yams to one of three quality grades: Extra Class, Class I, or Class II. The system uses two separately trained EfficientNetB0 models initialized with ImageNet weights. Each model combines a global-average-pooling head, batch normalization, dropout, and a task-specific classifier. The reported image collection contains 795 images, including 303 non-yam images and yam examples annotated with reference to the Philippine National Standard PNS/BAFPS 96:2010. Training uses image augmentation, class weighting, selective fine-tuning of the top 20 backbone layers, and early stopping. On the reported validation data, the binary stage achieved 93.67% accuracy, while the grading stage achieved 86.73% accuracy and a weighted F1-score of 0.87. Class II, Extra Class, and Class I obtained F1-scores of 0.90, 0.88, and 0.78, respectively. The results support the feasibility of hierarchical image classification for purple yam grading, while the lower Class I recall and need for independent evaluation motivate further validation. The reported results are validation estimates rather than evidence of field-level performance.**

***Index Terms*—purple yam, ube, agricultural grading, computer vision, EfficientNetB0, transfer learning, hierarchical classification**

I. Introduction

Agricultural grading supports consistent handling, marketing, and quality control across food supply chains. Root and tuber crops can be difficult to inspect at scale because their external appearance varies with cultivar, maturity, growing conditions, handling, and storage. In many settings, grading remains a manual visual task, which takes time and may produce inconsistent decisions across inspectors. Computer vision offers a way to support this work by processing images with repeatable decision rules and learned visual representations.

Purple yam, or ube (*Dioscorea alata*), is a valued food crop in the Philippines and other parts of Southeast Asia. Its characteristic color and culinary uses have contributed to demand for fresh tubers and ube-based products. However, automated quality grading of purple yam has received less attention than inspection tasks for several other crops. Prior studies have explored computer-vision-based carrot grading, root-crop identification, cassava disease classification, sweet-potato grading, and non-destructive yam quality assessment using hyperspectral imaging and machine learning [3]–[8]. These works demonstrate the broader potential of image-based and learning-based methods, but they do not establish a dedicated, standards-oriented pipeline for purple yam quality grades.

This paper presents YamScan, a hierarchical classification system intended to address that gap. Given an image, its first model predicts whether the image contains a yam. If accepted by this gate, the second model predicts one of three grades—Extra Class, Class I, or Class II—using labels informed by the Philippine National Standard (PNS/BAFPS 96:2010) for yam classification and grading [13]. This paper describes the EfficientNetB0 transfer-learning pipeline and reports validation metrics, class-level results, and a small sample-based integrated pipeline demonstration.

The contributions are threefold: (1) a structured image collection for purple yam and non-yam recognition; (2) a two-stage classification design that separates crop identification from grade assignment; and (3) an empirical report of validation performance, including per-class grading metrics and an explicit discussion of evaluation limitations. The paper does not claim that the system replaces trained graders or that it has been validated in operational farm or market conditions.

II. Related Work

A. Computer Vision for Agricultural Inspection

Convolutional neural networks (CNNs) learn image features directly from labeled examples and have been applied to crop disease detection, defect inspection, and product grading. Deng *et al.* combined computer vision and deep learning for online carrot defect detection and grading, illustrating a route toward automated sorting [3]. In root crops, Seethalakshmi *et al.* studied CNN-based cassava disease classification [5], while Sreekumar *et al.* examined post-harvest physiological deterioration in cassava using deep learning [12]. These tasks differ from the present work: disease classification and defect detection typically target pathology, whereas YamScan assigns market-oriented quality categories to purple yam.

Object-detection approaches have also been considered in agricultural systems. Clavo de Comer *et al.* applied YOLOv9 to identifying multiple root crops, including purple yam, cassava, potato, and taro [6]. Sweet-potato studies report

machine-vision-based grading and detection [7] and pruned attention-based models for quality detection [8]. These studies motivate automation in crop handling but differ in crop, target label, or model objective. Their reported performance cannot be compared directly with YamScan because datasets, class definitions, partitions, and evaluation protocols are not harmonized.

B. Transfer Learning and Quality Assessment

Transfer learning reuses representations learned from a larger pretraining dataset and adapts them to a smaller target dataset. It is useful where agricultural labels are costly to acquire and the image collection is modest. EfficientNet architectures scale network depth, width, and input resolution in a coordinated manner; their ImageNet-pretrained features offer a practical starting point for image classification. Agricultural work has also used ensembles of pretrained CNNs for potato-tuber disease classification [9] and EfficientNet-based networks for root-crop leaf disease recognition [10]. Semi-supervised hyperspectral food-anomaly detection provides another example of learning from non-visible image spectra, but its target and sensing modality differ from YamScan [11]. These applications support the feasibility of transfer learning but do not establish that one architecture is universally optimal for tuber grading.

Image-based systems observe only visible evidence. Adesokan *et al.* investigated hyperspectral imaging and machine learning for evaluating dry-matter content in raw yams, illustrating that quality assessment can extend beyond ordinary color imagery [4]. For YamScan, the target is visual classification according to grade labels; the system does not directly measure internal quality, composition, or properties that require destructive or specialized sensing.

C. Research Gap

Existing studies demonstrate useful methods for agricultural detection and grading, yet relatively little work focuses specifically on automated purple-yam quality grading. YamScan addresses this narrow gap with a staged image-classification design and a label scheme anchored to the Philippine standard. Its contribution is an initial feasibility study; stronger evidence requires a larger, independently partitioned dataset and evaluation across farms, harvest periods, devices, and operating environments.

III. Methodology

A. Study Design and Dataset

The work follows a developmental and experimental design: images were assembled and annotated, classification models were trained, and performance was summarized on held-out validation subsets. The reported collection comprises 795 images gathered from farms, agricultural partners, local markets, and field observations, including work with Cabalen Aeta Farmers and Partners in Porac, Pampanga. Images were captured using accessible devices such as smartphones and laptop cameras. The acquisition process sought variation in viewpoints, lighting, and backgrounds so that images would represent more than a single controlled setup.

TABLE I
Reported Dataset Inventory

Category	Images	Share of reported total
Extra Class	212	26.67%
Class I	119	14.97%
Class II	161	20.25%
Not yam	303	38.11%
Total	795	100.00%

The grade labels are Extra Class, Class I, and Class II, based on PNS/BAFPS 96:2010 [13]. This paper reports 212 Extra Class images, 119 Class I images, 161 Class II images, and 303 non-yam images, for a total of 795 images.

The standard distinguishes grades by whole-tuber condition, maturity, firmness, cleanliness, shape, and allowable extent of defects. Extra Class represents the highest visual quality, with stringent limits on defects. Class I permits limited minor defects while retaining good overall presentation. Class II includes tubers that fail the higher-grade requirements but satisfy baseline quality conditions. These distinctions define an ordered grading task, although the implemented model treats the three grades as mutually exclusive nominal classes rather than exploiting their ordinal relationship.

B. Image Processing and Data Splitting

Images are resized to 224×224 pixels to match the classification input pipeline. This paper describes standard resizing for validation and on-the-fly augmentation for training. Augmentations include horizontal and vertical flips, rotations of up to approximately 20 degrees, zoom of up to 20%, and translations of up to 10% in either spatial direction. Applying these transformations to training images increases variation in orientation, framing, and apparent scale without modifying the validation examples.

An 80/20 training–validation split with a fixed random seed of 42 is reported for both stages. The validation subsets are used to monitor accuracy and loss, select model weights through early stopping, and calculate classification reports and confusion matrices. Available documentation does not specify a separate, untouched test set, nor does it state whether images of the same physical tuber or acquisition session were kept together during partitioning. Consequently, the reported validation metrics should be interpreted as estimates under the described split, not as independent evidence of generalization to new farms or acquisition domains.

C. Hierarchical Classification Pipeline

YamScan uses two classifiers in sequence. Stage 1 is a binary gate that predicts *yam* or *not yam*. A positive yam prediction routes the image to Stage 2; a negative prediction terminates processing. Stage 2 assigns a positive image to Extra Class, Class I, or Class II. The threshold described for the binary output is 0.5, and the second stage selects the grade with the greatest softmax probability.

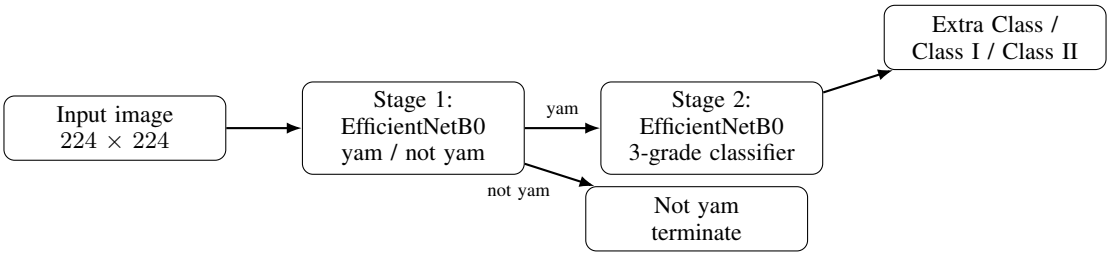


Fig. 1. YamScan's sequential image-classification flow. Stage 2 runs only after Stage 1 accepts an image as yam.

The hierarchy decomposes the overall decision into an object-validity decision followed by a quality-grade decision. This may simplify the grade classifier's input space, but it also introduces error propagation: a yam image rejected by Stage 1 cannot be recovered by Stage 2, while a false-positive non-yam image receives an unsupported grade. This paper reports the two stages separately and a demonstration using a small sample of images; it does not report a statistically powered end-to-end test set from which full-pipeline grade accuracy could be estimated.

D. EfficientNetB0 Models

Both stages use EfficientNetB0 initialized with ImageNet-pretrained weights. The convolutional backbone is followed by global average pooling, batch normalization, dropout with rate 0.3, and a task-specific dense output. Stage 1 uses one sigmoid unit with binary cross-entropy loss. Stage 2 uses three softmax outputs with categorical cross-entropy loss. The binary output probability is thresholded at 0.5; the multiclass prediction is selected by maximum softmax score.

Training proceeds in two phases. First, the pretrained backbone is frozen while the classification head is trained. Next, the top 20 layers of the backbone are unfrozen for selective fine-tuning with a lower learning rate. This paper reports a fine-tuning learning rate of 10^{-4}, dynamic class weights calculated from the training labels, and early stopping based on validation loss with patience of three epochs and restoration of the best weights. Class weighting is intended to reduce the effect of imbalanced label counts; it does not add new examples or remove the need to evaluate per-class behavior.

E. Evaluation Metrics

Performance is summarized using accuracy, precision, recall, F1-score, and confusion matrices. For a binary positive class, these quantities are

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{3}$$

$$F_1 = 2\frac{\text{Precision}\,\text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4}$$

For the three-grade task, precision, recall, and F1 are computed per class; macro averages weight classes equally, whereas weighted averages account for class support. Confusion matrices are used to inspect which labels are mistaken for one another. Since Class I recall is notably lower than the other class recalls, accuracy alone would obscure an important operational weakness.

F. Labeling Scope and Annotation Considerations

The category labels represent expert-informed visual grades, not direct measurements of internal composition or consumer preference. This paper describes assistance from the chairperson of Cabalen Aeta Farmers and Partners in categorizing the collected examples. It does not specify the number of independent annotators, a blinded adjudication procedure, inter-rater agreement, or a documented resolution process for borderline samples. Those details matter because model performance is bounded by the consistency of the reference labels. Two images of the same tuber may also differ substantially with viewing angle, illumination, or occlusion, while a single image may not show all defects needed for a confident grade. Accordingly, the learned prediction should be understood as a mapping from the available image to the assigned dataset label, not a complete inspection of the physical product.

This paper discusses ripeness, freshness, defects, and accepted or rejected yams as broader motivations. The implemented output labels documented for the grading model, however, are only Extra Class, Class I, and Class II, preceded by a yam/not-yam decision. No separate ripeness classifier, disease classifier, defect-localization module, or explicit reject/uncertain category is reported. This distinction defines the present system's scope and avoids implying that the classifier performs functions beyond the stated model outputs.

G. Training and Validation Procedure

The workflow first organizes the original folders into a binary data view for Stage 1 and a three-folder grade view for Stage 2. Stage 1 combines the three yam grade folders into the positive yam category and uses non-yam images as the negative category. Stage 2 uses only the three yam-grade categories. Each task is then split into training and validation subsets, and class weights are computed for the relevant training labels. This staging supports independent optimization of the two decisions while preserving a common image-processing approach.

During transfer learning, the frozen EfficientNetB0 backbone serves as a feature extractor while the newly attached

classifier head is optimized. Selective unfreezing then allows the upper backbone layers to adapt to the visual distribution in the target collection. The reported patience of three epochs and best-weight restoration reduce the risk of keeping weights from a later epoch with worse validation loss. This paper describes use of the Adam optimizer and a learning rate of 10^{-4} for fine-tuning, but does not provide a complete training configuration in the report: batch size, exact head-training schedule, software versions, hardware, random seeds for model initialization, and per-stage stopping epoch details are not all specified. These omissions limit exact reproducibility and should be addressed in a future technical release.

The validation split is used in multiple ways: epoch-level loss and accuracy monitoring, early stopping, and final metric reporting. This is common in small-data development, but means the validation set is not a fully independent final test after model selection. Repeated model adjustment based on these results can make the selected validation score optimistic. A more rigorous follow-up should lock model choices using training and validation data, then evaluate once on a held-out test collection gathered from disjoint tubers, capture sessions, and preferably locations. Repeated seeds or cross-validation can additionally characterize sensitivity to the particular partition, although neither substitutes for a genuinely external test.

H. Interpreting the Cascaded Decision

Let D denote the event that Stage 1 accepts an image as yam and let G denote the event that Stage 2 assigns the correct grade conditional on receiving a yam image. A correct final grade requires both decisions to succeed:

$$P(\text{correct final grade}) = P(D \mid \text{yam})\, P(G \mid D, \text{yam}). \quad (5)$$

This expression highlights why stand-alone stage metrics do not determine the complete pipeline's performance. The reported Stage 1 yam recall and Stage 2 accuracy are measured on separate validation subsets and cannot simply be multiplied to claim an end-to-end grade accuracy; the required joint conditional probabilities were not reported. An integrated test should apply both models to one independent, grade-labeled collection and count rejected yams, accepted non-yams, and each final grade error. It should also report the number of samples that reach each stage.

Threshold selection is another operational choice. The stated 0.5 threshold offers a simple binary rule, but it does not necessarily minimize the cost of either rejecting a valid yam or routing an irrelevant image to grading. Thresholds can be selected using validation data to meet an agreed sensitivity or workload target, then frozen before independent testing. Similarly, softmax outputs are model scores and should not automatically be interpreted as calibrated probabilities. Calibration checks and an optional manual-review band could reduce overconfident decisions, particularly for borderline images and Class I cases.

TABLE II
STAGE 1 VALIDATION RESULTS

Class	Precision	Recall	F_1	Support
Not yam	0.93	0.90	0.92	61
Yam	0.94	0.96	0.95	97
Overall accuracy		0.9367		158

IV. RESULTS

A. Stage 1: Yam–Not-Yam Classification

The refined binary model was trained for 12 epochs. Reported training accuracy increased from 62.99% in the first epoch to 92.44% in the twelfth, with a peak of 94.96% at epoch 11. Validation accuracy rose from 70.89% to 93.67%, reaching a reported peak of 94.94% at epoch 10. Training loss declined from 0.7215 to 0.1647, and validation loss declined from 0.5073 to 0.1474. The final-epoch value and peak value are distinct; the results below use the stated final validation accuracy unless otherwise noted.

The validation confusion matrix contains 158 images: 55 non-yam images were correctly rejected, 93 yam images were correctly accepted, six non-yam images were falsely accepted as yam, and four yam images were rejected. These counts imply high recall for the yam category and relatively few false-positive and false-negative decisions on this validation subset. This paper's classification report gives precision of 94% and recall of 96% for yam, and precision of 93% and recall of 90% for non-yam. Reported F1-scores are 0.95 for yam and 0.92 for non-yam. The reported validation support of 158 is approximately 19.9% of the 795-image collection, consistent with an 80/20 split up to partition rounding.

B. Stage 2: Three-Class Grading

The refined grading model was trained for 12 epochs. Training accuracy rose from 48.48% to 87.31%, with a maximum of 87.56% in epochs 9 and 11. Validation accuracy rose from 63.27% to 86.73%, reaching 87.76% in epochs 10 and 11. Training loss fell from 1.3244 to 0.3206, and validation loss reached 0.4827 at epoch 12. This paper reports an improvement from 82.65% before refinement to 86.73% after unfreezing the top 20 layers and applying class weights. This comparison is part of the reported training process; no repeated-seed results or uncertainty intervals are provided.

Table III summarizes the class-level validation report for 98 images. Class II achieved the highest precision (0.97) and an F1-score of 0.90. Extra Class had the highest recall (0.95) and F1-score of 0.88. Class I had the lowest recall (0.73) and F1-score (0.78), indicating that nearly one quarter of its validation examples were not retrieved as Class I. The macro-average F1-score was 0.86 and the weighted-average F1-score was 0.87. These results show useful separation for two categories but also establish Class I as the principal challenge.

This paper describes most grade confusions as occurring between neighboring categories, a plausible consequence of visually subtle boundaries between adjacent grades. However,

TABLE III
STAGE 2 VALIDATION CLASSIFICATION REPORT

Grade	Precision	Recall	F_1	Support
Class I	0.84	0.73	0.78	22
Class II	0.97	0.85	0.90	33
Extra Class	0.82	0.95	0.88	43
Macro average	–	–	0.86	98
Weighted average	–	–	0.87	98
Accuracy		0.8673		98

numerical entries of the Stage 2 confusion matrix were not available in the extracted report, so this paper does not reconstruct or infer individual off-diagonal counts. The class-level metrics provide the quantitative basis for the interpretation above.

C. Integrated Demonstration

This paper describes an end-to-end demonstration that selected at least one image from each of four folders (not yam, Class I, Class II, and Extra Class) and one additional random image. Each image was processed first by the binary model; Stage 2 was skipped when the prediction was not yam. The output displayed class folder, Stage 1 prediction and confidence, and, when applicable, Stage 2 grade and confidence. This test illustrates how the two independently trained models are connected. Because it covers only a handful of selected images and no aggregate outcomes are reported, it is a functional demonstration rather than an independent evaluation of integrated system accuracy.

V. DISCUSSION

The reported results suggest that transfer learning with EfficientNetB0 can support an initial purple-yam image classification system. The Stage 1 model attained 93.67% validation accuracy and F1-scores above 0.9 for both classes. In the grading stage, a weighted F1-score of 0.87 indicates promising aggregate performance on the validation split. Importantly, class-specific results make the system's uneven behavior visible: Class II precision is high, Extra Class recall is high, and Class I recall is materially lower. A grading assistant should expose these distinctions rather than report a single accuracy figure alone.

The hierarchical arrangement has an interpretable operational logic. A separate gate can screen out non-yam images before grade estimation, reducing the number of irrelevant images handled by the grading model. It can also concentrate the second stage on the intended crop. This decomposition is not automatically superior to a single four-class classifier, however. Stage 1 errors propagate, and two models create separate calibration and maintenance requirements. A future comparison should evaluate a flat classifier and alternative ordinal or cost-sensitive approaches on the same image partitions. In particular, mistakes between adjacent grades may have different consequences from mistakes that cross multiple grades, and the current nominal cross-entropy objective does not encode that distinction.

Class imbalance was addressed with computed class weights and training-only augmentation. These measures are appropriate for the unequal class counts reported, but cannot compensate for incomplete coverage of cultivars, growing regions, tuber conditions, cameras, or seasons. The lower Class I recall may reflect limited support, overlapping visual appearances, annotation ambiguity, or other dataset properties; the current evidence cannot isolate a cause. The next iteration should review Class I labels with qualified graders, collect additional examples near grade boundaries, and examine the confusion matrix alongside image-level errors.

Several evidence limitations constrain interpretation. First, no independent test set is described, and validation data are used during early stopping and model selection. Second, grouping by physical tuber, farmer, capture session, or location before partitioning is not documented; if related images appear in both subsets, validation performance may be optimistic. Third, sample-based end-to-end testing does not provide a statistically meaningful estimate of system-level accuracy. Finally, performance on images cannot establish inspection-time savings, reduced post-harvest losses, or agreement with professional graders. These are outcomes for future field evaluation, not outcomes evaluated in this paper.

The reported validation numbers also should not be compared as a ranking against other crop studies. Prior work covers different crops, target categories, image sensors, dataset sizes, and train/test protocols [3], [5]–[9]. A valid comparative study would require common data, agreed grade definitions, matched splits, and consistent metrics. YamScan's present result is best interpreted as evidence of feasibility within the collected dataset.

A. Implications for Agricultural Use

If validated further, YamScan could act as a decision-support aid: it can return a preliminary category from an image and make an initial visual sort more consistent. Its likely role is complementary rather than substitutive. A human operator can inspect uncertain predictions, confirm that the image contains one clearly visible tuber, and make the final grading decision under the applicable standard. This workflow is especially relevant where the model's category confidence is weak or where the photographed view obscures a relevant surface feature. The evaluation did not measure operator agreement, system throughput, time saved, or changes in post-harvest loss; these outcomes should be treated as evaluation questions rather than established benefits.

Practical deployment also depends on the acquisition protocol. Camera distance, focus, background, lighting, tuber cleanliness, number of objects in frame, and partial occlusion can all affect the visual evidence available to a classifier. Although the dataset was collected under varied conditions, the paper does not quantify coverage or performance by device, lighting, collection site, or viewpoint. A controlled capture guide, repeatable framing, and examples of unsupported inputs would help users produce images closer to the training distribution. In subsequent field trials, subgroup metrics should be

reported for acquisition settings and collection sites to identify domain shifts that aggregate validation scores can hide.

A production-oriented evaluation should predefine the unit of prediction and the decision costs. If a photograph contains multiple tubers, the current image-level classifier does not provide localized boxes or individual grades; it should not be assumed to separate several objects in one frame. If the system is intended to grade one tuber at a time, the interface should communicate that constraint and provide a clear path for invalid or ambiguous images. Reference labels should be assigned independently of model predictions, and disagreements should be adjudicated by qualified graders. Recording model version, image provenance, prediction, confidence, and final human decision would support auditability and continuous quality review.

B. Limitations and Threats to Validity

The sample size is modest for learning robust visual distinctions across a heterogeneous agricultural product, and the class distribution is uneven. In particular, Class I has 119 reported images overall and 22 validation examples. With this support, a small number of different predictions could materially change its reported recall and F1-score. Confidence intervals or repeated split estimates were not reported, so the precision of the point estimates is unknown. The class distribution is uneven, and a larger, more diverse image collection is needed before replication or broader generalization can be established.

The provenance and independence of images are not fully documented in the report. Multiple views of one tuber, images captured in one session, or images from the same collection site may share visual cues. If related images cross the training and validation boundary, the validation set may be easier than genuinely new data. Validation performance is also used during model refinement and early stopping, and does not identify an untouched test set. Together, these properties constrain external validity and make the reported values preliminary.

There are also construct-validity limits. The labels are based on a visual grading standard, but the paper does not report annotation agreement or explain how ambiguous specimens were handled. A photograph may not reveal all the properties required for a physical grade, particularly when defects are hidden or internal. As a result, an image classifier can reproduce visual patterns associated with the dataset labels without necessarily reproducing the full inspection process performed by a human grader. The model's output should therefore be phrased as an image-based grade prediction and checked against qualified assessment in any practical trial.

Finally, comparisons with prior work are contextual only. Published studies on carrots, cassava, potatoes, and sweet potatoes use different sensors, collection sizes, label sets, and evaluation splits. A larger accuracy in another paper does not by itself establish superiority or inferiority. A fair benchmark would use shared test images and clearly defined targets, while a standards-compliance study would additionally compare automated decisions against adjudicated PNS labels. These limitations suggest a staged evidence program: correct data inventory, audit annotation, create a grouped independent split, validate the cascade, and only then measure usability and economics in the field.

VI. Conclusion and Future Work

This paper presented YamScan, a two-stage EfficientNetB0 pipeline for identifying yam images and assigning purple yam images to Extra Class, Class I, or Class II. On the reported validation subsets, Stage 1 achieved 93.67% accuracy, while Stage 2 achieved 86.73% accuracy and a weighted F1-score of 0.87. Per-class results were strongest for Class II and Extra Class, while Class I recall of 0.73 identifies a clear target for improvement. The work demonstrates the feasibility of transfer learning for standards-informed visual grading, but its validation results do not yet establish generalization or operational effectiveness.

Future work should first reconcile the dataset inventory and verify labels against the applicable standard with qualified graders. The collection should be expanded across farms, seasons, locations, devices, cultivars, backgrounds, and lighting conditions. Partitioning should be performed at the tuber or acquisition-session level where repeated views exist, followed by an independent test set that is not used for model selection. Evaluation should report per-class confusion matrices, precision, recall, F1-score, confidence intervals, and end-to-end pipeline results.

Further experiments can compare EfficientNetB0 with alternative CNN backbones, a single four-class model, class-balanced sampling, ordinal classification, and cost-sensitive learning. Confidence calibration and an abstention option for uncertain predictions may be valuable in a human-in-the-loop grading workflow. Finally, field trials with farmers and trained agricultural personnel should measure agreement with reference graders, throughput, usability, and failure cases under real handling conditions. Such evidence is necessary before YamScan can be recommended as a practical grading tool.

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